BioTilt - Power supply wiring diagrams

Block diagrams for the three power options in hardware/power-supply.md. Option C is the recommended one.

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Legend

- Outside the device: supplied by the user (charger, adapter, cable)
- Input module: where power enters the enclosure
- Protection / switching
- Dashed border = optional part
- Battery-related part (option A only)
- BioTilt electronics (same in every option)
- Consumption measurement (page 5)

- + supply (9 V rail)
- GND (common ground)
- Logic signals

Crossing wires with a hop are NOT connected. A dot marks a junction.

Key points

No 5 V step-down is needed. The Arduino Nano ESP32 accepts 6-21 V on VIN and has its own regulator. The same 9 V rail feeds the A4988 VMOT (8-35 V) and the Nano VIN. This is how the prototype already runs from a 9 V supply.

Mains (220 V) never enters the device. In options B and C the charger/adapter stays outside the incubator and only a low-voltage cable passes the door seal.

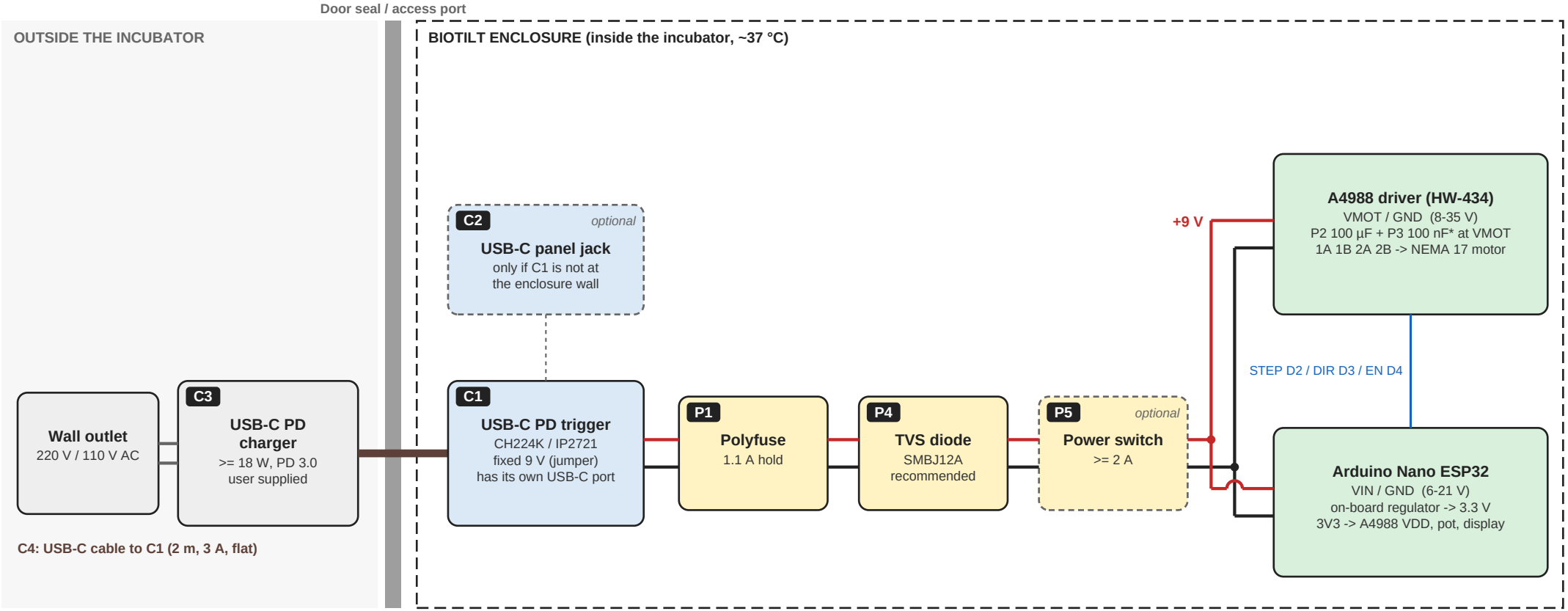
Capacitors at VMOT. P2 (100 µF electrolytic) absorbs slow, large voltage spikes from cable inductance and motor current, and protects the A4988 from going over 35 V. P3 (100 nF ceramic) filters high-frequency switching noise. *Most HW-434 / Pololu-style boards already include a ceramic capacitor on VMOT, so P3 is optional.

Common ground. The input GND, both A4988 GND pins and the Nano GND must be connected together.

Never connect or disconnect the motor while the driver is powered.

Option C - USB-C Power Delivery at 9 V (recommended)

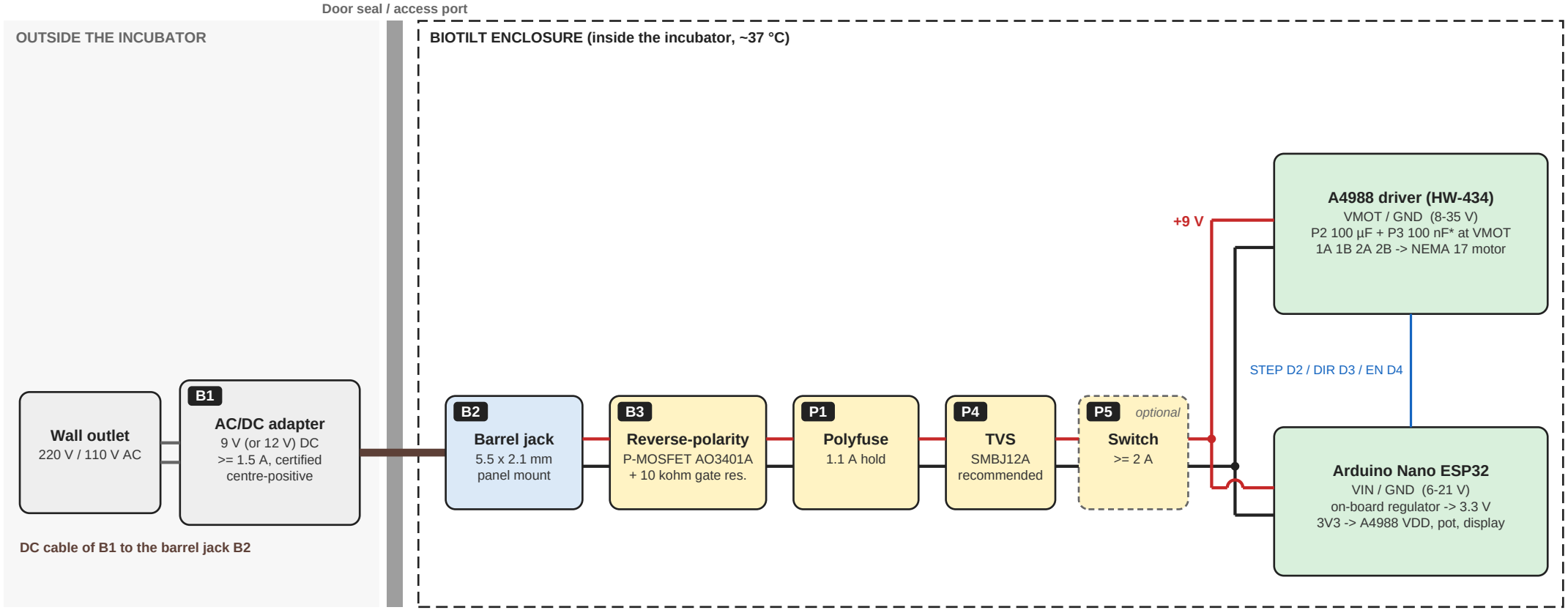
Any USB-C PD charger >= 18 W offers 9 V. Inside the device there is only ONE input module: the PD trigger board (C1).



ID	Part	What it does	Where	Required
C3	USB-C PD charger	Converts mains to USB-C; negotiates 9 V with C1	Outside (wall)	Yes (user supplied)
C4	USB-C to USB-C cable	Carries 9 V through the door seal	Through the seal	Yes (user supplied)
C1	USB-C PD trigger module	Asks the charger for 9 V and outputs it on + / - pads. **The only module in the device.**	Enclosure wall	Yes
C2	USB-C panel connector	Extension so the port sits on the wall if C1 is mounted further inside	Enclosure wall	Optional
P1	Polyfuse 1.1 A	Cuts current on a short circuit, resets by itself	Enclosure	Yes
P4	TVS diode SMBJ12A	Clamps input spikes below the Nano 21 V limit (see protections.md)	Enclosure	Recommended
P5	Power switch	On/off	Enclosure wall	Optional
P2 / P3	100 µF / 100 nF capacitors	Stabilise VMOT at the A4988 (see page 1)	At the A4988	P2 yes, P3 optional

Option B - 9 V mains adapter (barrel jack)

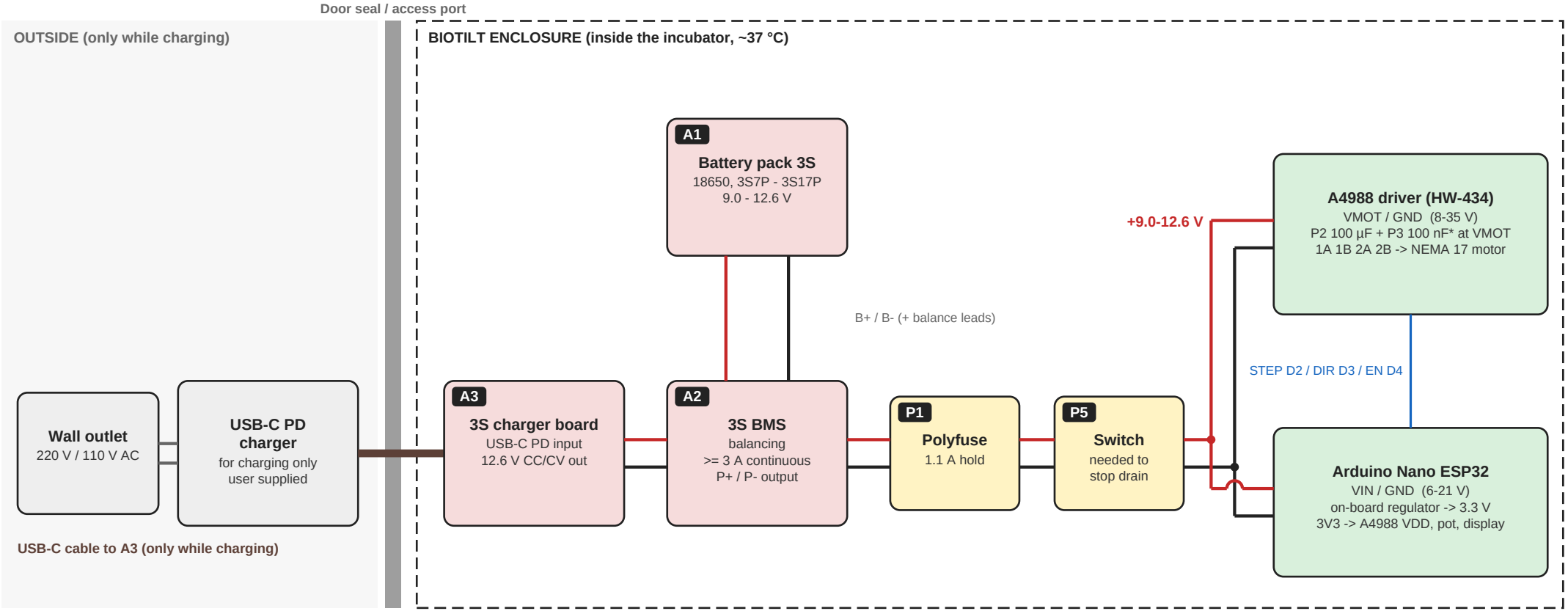
This is how the prototype runs today. Same 9 V rail as option C. Only the input module changes.



ID	Part	What it does	Where	Required
B1	AC/DC wall adapter 9 V	Converts mains to 9 V DC. Use 12 V for more margin above the A4988 8 V minimum	Outside (wall)	Yes
B2	Panel barrel jack 5.5 x 2.1 mm	Input connector on the enclosure wall	Enclosure wall	Yes
B3	P-MOSFET (AO3401A) + 10 kohm	Blocks current if the adapter has the wrong polarity. Drops much less voltage than a diode, which keeps VMOT above 8 V. Gate rating allows inputs up to 12 V	Enclosure	Yes
P1	Polyfuse 1.1 A	Cuts current on a short circuit, resets by itself	Enclosure	Yes
P4	TVS diode SMBJ12A	Clamps input spikes below the Nano 21 V limit (see protections.md)	Enclosure	Recommended
P5	Power switch	On/off	Enclosure wall	Optional
P2 / P3	100 µF / 100 nF capacitors	Stabilise VMOT at the A4988 (see page 1)	At the A4988	P2 yes, P3 optional

Option A - Rechargeable 3S Li-ion battery (reference only)

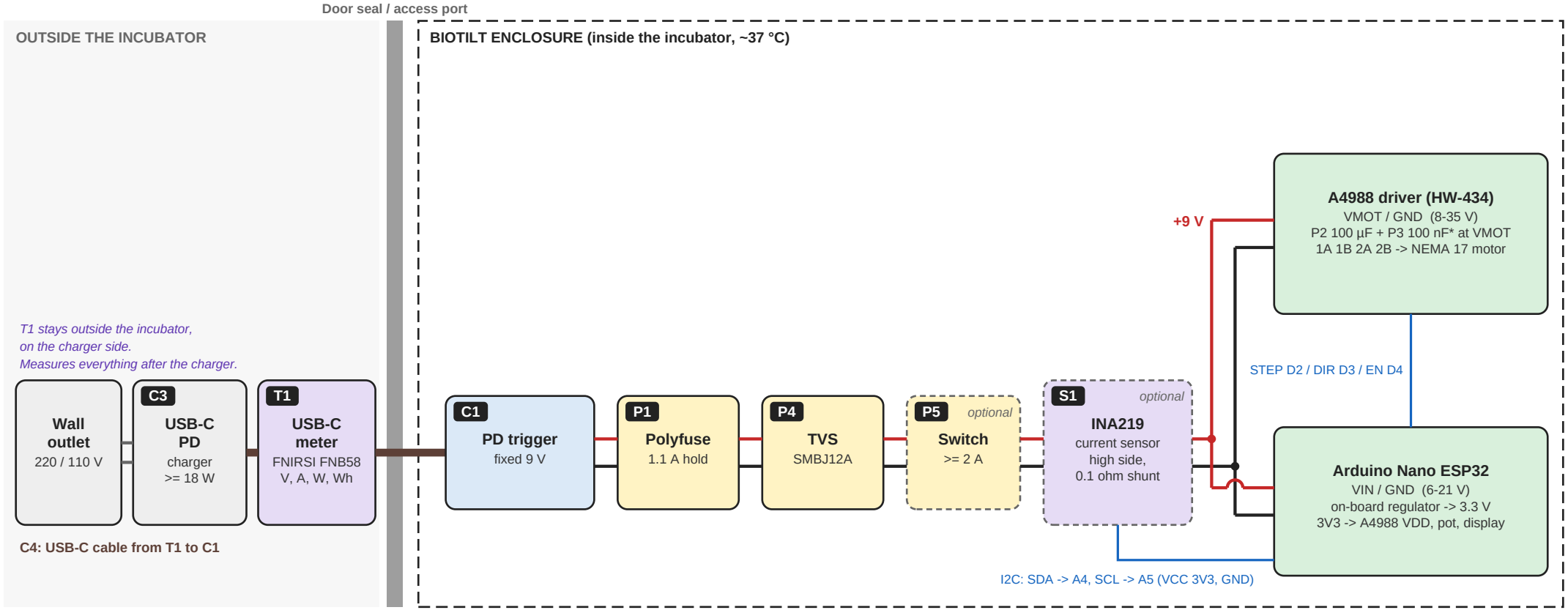
Not recommended for 1-week runs: ~550 Wh needed with the current motor (3S17P, ~51 cells). See power-supply.md section 3.



ID	Part	What it does	Where	Required
A1	Li-ion pack 3S (18650)	Stores energy. 3S = 10.8 V nominal, 9.0-12.6 V range: within A4988 and Nano limits, so **no step-down is needed**	Enclosure	Yes
A2	3S BMS with balancing	Protects cells (over/under-voltage, over-current) and balances them	Enclosure	Yes
A3	3S charger board (USB-C PD in)	Charges the pack at 12.6 V CC/CV from a USB-C charger	Enclosure wall	Yes
P1	Polyfuse 1.1 A	Short-circuit protection on the load side	Enclosure	Yes
P5	Power switch	Disconnects the load so the pack does not drain in storage	Enclosure wall	Yes
P2 / P3	100 µF / 100 nF capacitors	Stabilise VMOT at the A4988 (see page 1)	At the A4988	P2 yes, P3 optional

Consumption measurement (shown on option C)

T1 measures the real total consumption from outside, with no changes to the device. S1 is an optional internal sensor for logging from the firmware.



ID	Part	What it does	Where	Required
T1	Inline USB-C power meter (FNIRSI FNB58; alt. POWER-Z KM003C)	Shows live V / A / W plus accumulated Wh, mAh and time. Average power = Wh / hours. The screen does NOT store a time series: for a curve, keep it connected to a PC (CSV export) or the phone app for the whole run. Must pass PD negotiation through	External, between C3 and C4	Yes, for validation (one per project)
S1	INA219 current sensor module (alt. INA226 for 36 V / higher precision)	Measures V, A and W of the 9 V rail and sends them to the Nano over I2C (A4/A5, the pins left free for I2C). Placed before the star point it measures the TOTAL (motor + Nano). The firmware can log a time series (Serial, flash or microSD) without a PC	Inside, before the star point	Optional (future)
-	What to expect	The A4988 holds coil current almost constant, so consumption will be nearly flat (~0.3 A at 9 V estimated). Useful to confirm the power budget and detect faults (motor disconnected, wrong voltage). It does NOT detect a stalled motor	-	-